

Bootstrapping Risk-Bearing Capital: Solver Economics for Coordinated Liquidation in Onchain Perpetual Markets

Nihar Shah

Independent Researcher
nehharshah@gmail.com

Keywords:

Solver Economics, Liquidation, Batch Auction, Bootstrapping, DeFi, Perpetual Futures.

Abstract:

Coordinated batch liquidation mechanisms such as BALI can reduce cascade depth in onchain perpetual markets, but their effectiveness depends on whether rational solvers participate. This paper addresses the participation bottleneck with four contributions: (1) participation conditions under a cost model with dual-channel adverse selection, stress-dependent slippage, and multi-hour inventory risk, robust to $\beta \in [2, 8]$; (2) a concrete batch auction architecture specifying stress detection, coverage gating, sealed-bid clearing, settlement, and fallback logic; (3) a two-phase bootstrapping model (shadow mode $\rightarrow$ activation) resolving the paradox that BALI harms at low coverage yet must cold-start from zero, at total cost $\sim \$220K$; and (4) stress simulations on 3 exchanges showing 70%+ coverage eliminates cascades ($t = -21.1$, $p < 0.001$) while coverage below 50% increases bad debt 20–100 $\times$. Correlated default analysis shows maintaining 70% effective coverage requires diversification ($\rho_c \leq 0.15$, nominal 88%).

1 INTRODUCTION

Onchain perpetual exchanges process billions of dollars in daily liquidation volume through sequential, first-come-first-served mechanisms. When markets move sharply, these sequential pipelines become pro-cyclical: each forced liquidation widens market impact, degrades equity for other accounts, and triggers additional liquidations in a self-reinforcing cascade (Shah, 2026). The October 2025 crash demonstrated the consequences: \$10 billion liquidated on Hyperliquid in 24 hours, 40+ auto-deleveraging (ADL) events in 10 minutes, dYdX offline for 8 hours, and \$45–52 million in excess losses (CoinDesk Research, 2025; dYdX Foundation, 2025).

Prior work on batch liquidation (BALI) (Shah, 2026) showed that coordinated clearing can bound cascade depth ($D_t \leq 1$) under sufficient auction coverage, but identified—without resolving—the binding constraint: participation. This paper addresses the participation bottleneck with four contributions:

1. Participation conditions under a cost model with dual-channel adverse selection, stress-

dependent slippage, and multi-hour inventory risk, with sensitivity analysis showing robustness to parameter choices.

2. A concrete batch auction architecture specifying stress detection, coverage gating, sealed-bid clearing, on-chain settlement, and automatic fallback to continuous liquidation (Section 4).
3. A two-phase bootstrapping model—shadow mode followed by activation—resolving the paradox that BALI harms exchanges at low coverage yet requires a low-coverage cold-start period.
4. Stress simulations on 3 exchanges identifying both the coverage threshold for cascade elimination (70%) and the minimum viable coverage below which BALI is harmful ($< 50\%$).

Our central finding: solver bootstrapping is economically tractable (total cost $\sim \$220K$), but the mechanism’s binary nature (helpful above 70%, harmful below 50%) makes deployment harder than the economics suggest.

2 BACKGROUND AND MOTIVATION

2.1 Current Keeper Architectures

Vault-backstop (Hyperliquid). A three-stage waterfall: market orders, then HLP vault absorption, then ADL (Hyperliquid Labs, 2024). During the March 2025 JELLY crisis, exploited for \$13.5M (CoinDesk, 2025).

Validator daemon (dYdX v4). A liquidation daemon on each validator node (dYdX Foundation, 2024). October 2025 crash triggered an 8+ hour chain halt (dYdX Foundation, 2025).

Open MEV race (Aave-style). Highly concentrated: 2–3 bots capture most opportunities; capture probability collapses above 35ms latency (Perez et al., 2021; Qin et al., 2021; Enrico, 2025).

2.2 Precedents

MakerDAO (Black Thursday, 2020). Gas spikes prevented bidding; one bot won \$8.32M for zero DAI; \$4.5M bad debt (MakerDAO, 2020). CoW Protocol. Most mature batch solver market (KYC, competitions, token subsidies) (CoW Protocol, 2024), but handles ordinary swaps, not toxic liquidation flow.

2.3 The ADL Trilemma

ADL faces a fundamental trilemma: no policy simultaneously satisfies solvency, revenue, and fairness (Kulkarni et al., 2025). Production ADL achieves ~50% of the optimality bound (Kulkarni et al., 2026). Pre-ADL coordination relaxes this trilemma.

3 SOLVER PARTICIPATION GAME

3.1 Profit Function

Consider a solver j entering a BALI batch with total size Q and hourly volatility σ . With n solvers, each receives $q_j = Q/n$ under symmetric

Table 1: Equilibrium solvers and profit ($\sigma=0.015$, no subsidy).

Q	n^*	Profit	Viable?
\$10K	1	\$2.90	Marginal
\$50K	20	\$0.65	Marginal
\$100K	20	\$11.82	Yes
\$500K	20	\$195	Strong
\$1M	20	\$570	Strong
\$10M	20	\$18,611	Very strong

pro-rata allocation. Per-solver profit:

$$\Pi_j = \underbrace{(1-f_{\text{tox}}) \frac{(1-\gamma)\lambda Q^\alpha}{Q}}_{\text{discount}} q_j + \underbrace{s q_j}_{\text{subsidy}} - \underbrace{\eta\sigma\phi\psi q_j}_{\text{hedge}} - \underbrace{\frac{1}{2}\sigma^2 q_j \tau}_{\text{inv. risk}} - \underbrace{\rho q_j}_{\text{capital}} - \underbrace{g}_{\text{gas}}, \quad (1)$$

where $\lambda|q|^\alpha$ is the convex impact function from (Shah, 2026) and $\gamma=0.65$. Three cost terms extend prior work:

Execution slippage: $\phi(\sigma) = 1 + 3\max(0, \sigma/0.010 - 1)$. At $\sigma = 0.015$, $\phi = 2.5$; at $\sigma = 0.05$, $\phi = 13$.

Inventory risk: $\frac{1}{2}\sigma^2 q_j \tau$ with $\tau = 2$ hours, reflecting 1–4 hour hedge execution during stress.

Dual-channel adverse selection: (i) f_{tox} reduces the effective discount by $(1-f_{\text{tox}})$; (ii) amplifies hedge cost by $\psi = 1 + \beta f_{\text{tox}}^2$ ($\beta = 4$). This ensures adverse selection binds at practical batch sizes.

3.2 Participation Threshold

Condition 1 (Participation). A solver enters if $\Pi \geq 0$. The minimum batch size for n solvers:

$$Q^*(n, \sigma) \approx \left(\frac{ng}{A\lambda + s - C} \right)^{1/\alpha}, \quad (2)$$

where $A = (1-\gamma)(1-f_{\text{tox}})$ and $C = \eta\sigma\phi\psi + \frac{1}{2}\sigma^2\tau + \rho$. For $n=5$ at $\sigma=0.015$: $Q^* \approx 19,000$.

Condition 2 (Equilibrium). n^* is the largest n with $\Pi \geq 0$, capped at 20.

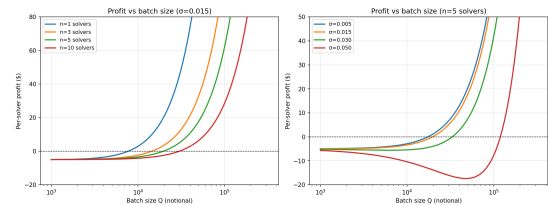


Figure 1: Solver profit. Left: by solver count. Right: by volatility ($n=5$).

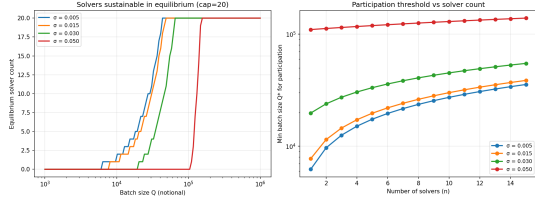


Figure 2: Left: equilibrium solver count vs batch size. Right: Q^* vs solver count.

3.3 Comparative Statics

- Hedging cost (η): Higher η increases Q^* (Figure 3).
- Volatility: At $\sigma = 0.03$, $\phi = 7$, sharply raising Q^* .
- Subsidy: 3 bps substantially reduces the 5-solver threshold (Figure 4).

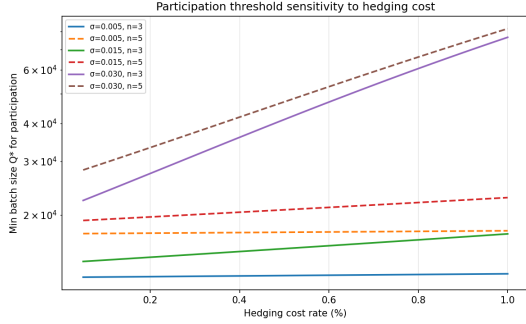


Figure 3: Participation threshold Q^* vs hedging cost rate.

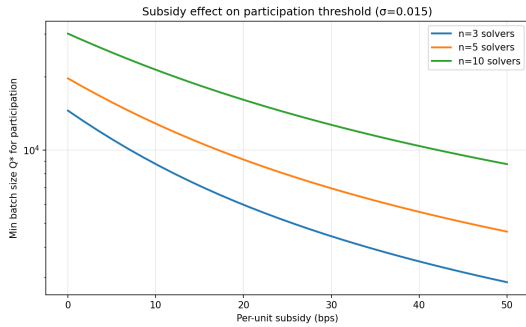


Figure 4: Per-unit subsidy reduces minimum batch size for participation.

3.4 Adverse Selection

Table 2 reports f_{tox}^* at zero profit ($n=5$). At $\sigma=0.03$ with \$50K batches, solvers exit at 24% toxicity.

Table 2: Critical toxic fraction f_{tox}^* ($n=5$, $\beta=4$).

Q	$\sigma=0.005$	$\sigma=0.015$	$\sigma=0.030$
\$50K	0.75	0.62	0.24
\$100K	0.89	0.78	0.46
\$500K	0.97	0.92	0.72
\$1M	0.98	0.94	0.79

Table 3: f_{tox}^* sensitivity to β ($n=5$).

Scenario	$\beta = 2$	$\beta = 4$	$\beta = 8$
$Q = \$50K$, $\sigma = 0.015$	0.64	0.62	0.59
$Q = \$50K$, $\sigma = 0.030$	0.26	0.24	0.22
$Q = \$100K$, $\sigma = 0.015$	0.80	0.78	0.74
$Q = \$100K$, $\sigma = 0.030$	0.50	0.46	0.41

Sensitivity to β : Varying β from 2 to 8 shifts critical fractions by $\sim 20\%$ (Table 3), confirming the qualitative finding is robust to the specific functional form.

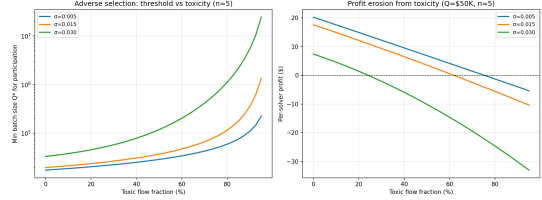


Figure 5: Left: Q^* vs toxic flow fraction. Right: profit erosion at $Q=\$50K$.

3.5 Correlated Default

If k of n solvers default mid-batch, surviving coverage degrades. With stress boost p_c and surviving fraction $1 - p_{\text{stress}}$, the required nominal coverage for 70% effective is $0.70/(1 - p_{\text{stress}})$:

Design requirement: diversification must keep $p_c \leq 0.15$ (nominal 88%). At $p_c = 0.30$, the requirement exceeds 100%—infeasible. The architecture (Section 4) enforces this via solver registry constraints.

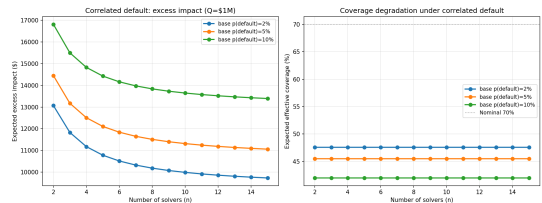


Figure 6: Left: excess impact from defaults ($Q=\$1M$). Right: coverage degradation (nominal 70%); at $p_c=0.3$, effective is 45%.

Table 4: Required nominal coverage for 70% effective.

ρ_c	p_{stress}	Surv.	Nominal	Feasible?
0.10	0.15	0.85	82%	Yes
0.15	0.20	0.80	88%	Yes
0.20	0.25	0.75	93%	Tight
0.30	0.35	0.65	108%	No

4 PROPOSED ARCHITECTURE

We specify the batch auction protocol at sufficient detail for security analysis. The architecture has five components: stress detection, coverage gating, batch auction, settlement, and failure handling. Figure 7 shows the system flow; Algorithm 1 formalizes the protocol.

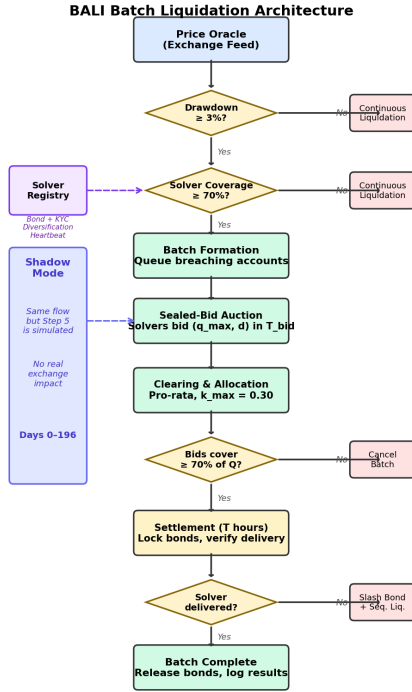


Figure 7: BALI architecture. Decision diamonds are coverage gates and undersubscription checks; red boxes are fallback to continuous liquidation. Shadow mode (left, days 0–196) runs the same flow with simulated settlement.

Algorithm 1: BALI Batch Liquidation Protocol

Input: Price feed P_t , account set $\mathcal{A}$, solver registry $\mathcal{S}$

```

// 1. Stress Detection
dd_t ← (max_{s ∈ [t-w, t]} P_s - P_t) / max_{s ∈ [t-w, t]} P_s
if dd_t < θ_stress then
    | return continuous liquidation
end

// 2. Coverage Gate
n_active ← |{s ∈ S :
    bonded(s) ∧ heartbeat(s, t)}|
c_eff ← n_active / n_target
if c_eff < θ_cov then
    | return continuous liquidation // fail
    | back
end

// 3. Batch Formation
B ← {a ∈ A : margin(a, P_t) < m_maint}
Q ← Σ_{a ∈ B} |q_a|

// 4. Sealed-Bid Auction (T_bid window)
Broadcast (B, Q, P_t) to S
Collect bids {(s_i, q_i^max, d_i)} within T_bid
Sort by discount d_i ascending
Allocate pro-rata, subject to q_i ≤ κ_max Q
Q_covered ← Σ_i q_i
if Q_covered < θ_cov · Q then
    | Cancel batch; revert to continuous
end

// 5. Settlement
Lock solver bonds for allocated positions
Solvers settle within τ_settle
Slash bond for non-delivery
Release remaining positions to continuous pipeline
  
```

4.1 Stress Detection

The protocol monitors a rolling drawdown dd_t over window w (e.g., 2 hours). When $dd_t \geq \theta_{\text{stress}}$ (e.g., 3%), the system evaluates whether to form a batch. The oracle is the exchange’s own price feed—the same feed used for margin calculation—so no external dependency is introduced. False positives (triggering on noise) are mitigated by the coverage gate: if solvers are not ready, the system falls back silently.

4.2 Coverage Gate

Before forming a batch, the protocol checks real-time solver availability. Each registered

solver maintains a heartbeat (signed message every T_{hb} seconds). The effective coverage $c_{\text{eff}} = n_{\text{active}}/n_{\text{target}}$ must exceed $\theta_{\text{cov}} = 0.70$. If not, the batch is not formed and accounts are liquidated continuously—the exchange never enters the dangerous low-coverage regime (Section 6.4).

This gate also handles mid-crisis coverage drops: if solver defaults push c_{eff} below θ_{cov} between batches, subsequent batches are suppressed and the system reverts to continuous liquidation automatically.

4.3 Sealed-Bid Auction

Once a batch is formed, the coordinator broadcasts the batch manifest to all active solvers. Each solver submits a sealed bid (q_i^{max}, d_i) : the maximum notional they will absorb and the minimum discount they require. The bid window T_{bid} is short (e.g., 60–300 seconds) to limit stale-price risk.

Clearing rule (discriminatory procurement): The exchange is procuring liquidation services. Bids are sorted by required discount d_i ascending. Each winning solver receives positions at their own bid discount (pay-as-bid), not a uniform clearing price. Allocation is pro-rata among accepted bids, subject to $\kappa_{\text{max}} = 0.30$ (minimum 4 solvers per fully-covered batch). Under discriminatory pricing, solvers may shade bids upward; however, the exchange’s credible fallback to continuous liquidation (which yields zero discount to all solvers) caps the extractable rent: any solver bidding above the sequential-impact discount gains nothing.

Stale-price risk: The bid window T_{bid} trades off participation (longer windows allow more solvers to respond) against stale-price risk (during a 10% crash, prices can move 1–2% in 5 minutes). Solvers who bid early face adverse price movement before settlement. Shorter windows (60s) reduce this risk but may exclude slower solvers; the optimal T_{bid} is deployment-specific.

Undersubscription: If total bid volume covers less than $\theta_{\text{cov}} \cdot Q$, the batch is cancelled and accounts revert to continuous liquidation, preventing the worst-of-both-worlds outcome.

4.4 Settlement and Slashing

Winning solvers receive allocated positions and have τ_{settle} (e.g., 2–4 hours) to hedge or close them. Their bond B (posted at registration) is locked for the duration.

Delivery verification: After τ_{settle} , the protocol checks whether each solver’s net position matches their allocation. Non-delivery triggers bond slashing: the slashed amount covers the additional impact cost of sequentially liquidating the defaulted share. Specifically, the slash for solver i defaulting on allocation q_i is $\text{slash}_i = \min(B_i, \lambda |q_i|^\alpha \cdot 1.5)$, reflecting the $1.5\times$ impact penalty for forced sequential liquidation of the defaulted share.

Partial clearing: If k solvers default post-allocation, the remaining positions ($\sum_{\text{defaulted}} q_i$) enter the continuous liquidation pipeline. The slash revenue partially offsets the additional impact cost.

4.5 Solver Registry and Diversification

The solver registry enforces diversification requirements that keep $p_c \leq 0.15$ (Table 4):

- No two solvers may share the same infrastructure provider (cloud region, co-location facility)
- Maximum 3 solvers from the same legal jurisdiction
- Minimum bond B_{min} scales with batch size capacity

Registration requires: (i) posting bond $B \geq B_{\text{min}}$, (ii) passing a shadow-mode qualification period (minimum 30 days of successful shadow batch processing), and (iii) demonstrating independent infrastructure.

4.6 Bootstrapping Integration

During the shadow-mode phase (Section 5), the architecture operates identically except that Step 5 (settlement) is simulated: solvers report what they would have done, positions are not actually transferred, and the exchange continues with continuous liquidation. The coverage gate (θ_{cov}) must be met in shadow mode before the protocol transitions to live mode. This means activation is not a manual decision—it is gated by the architecture’s own coverage check.

5 BOOTSTRAPPING MECHANISM DESIGN

5.1 The Activation Paradox

Section 6.4 shows BALI at <50% coverage increases bad debt. But bootstrapping begins at 0%. The architecture’s coverage gate (Section 4) resolves this structurally: BALI cannot activate until $c_{\text{eff}} \geq 0.70$. The question becomes: how do we fund solver infrastructure development during the pre-activation period?

5.2 Shadow Mode (Days 0–196)

BALI runs in parallel with continuous liquidation (Algorithm 1, shadow variant). Solvers process test batches against real market data, earning:

- Per-unit subsidy on shadow batches: $s(t) = 0.003 \cdot e^{-0.008t}$
- Standby stipend: \$50/day per solver (~\$1,500/month), calibrated to cover baseline infrastructure costs: cloud VM with GPU (\$200–600/month), market data feeds (\$200–500/month), monitoring and alerting (\$50–100/month), and hedging venue API access (\$100–300/month). This exceeds raw infrastructure costs but accounts for engineering time spent maintaining the solver codebase against exchange API changes.

Shadow mode serves three purposes: infrastructure development, performance validation (accumulating data to confirm the 70% threshold), and solver qualification for the registry (Section 4). We note that the bootstrap simulation applies the full cost model (hedging, inventory risk) to shadow batches where no real positions change hands. This makes the cost estimate conservative: shadow-mode solvers incur analysis costs but not risk-bearing costs, so actual shadow-mode profitability is higher than modeled, and activation would occur earlier. The \$220K figure should be treated as an upper bound.

5.3 Activation and Self-Sustaining Operation

When 14 of 20 registered solvers are active and bonded ($c_{\text{eff}} \geq 0.70$), the coverage gate passes and BALI goes live (day 196 in simulation). Subsidy continues declining; standby stipend ends. By day 339, subsidy is negligible and

solvers operate on stress-event economics: ~7 events/year, each producing 3–5 batches at $Q \geq \$500K$ ($\geq \$195/\text{solver}/\text{batch}$), yielding ~\$5,000–\$10,000/year from stress events alone.

5.4 Cost

Total: ~\$220,000 over 12 months. Readiness phase: \$178K (shadow subsidies + standby stipends). Active phase: \$43K (declining per-unit subsidy). This is ~0.5% of a single major stress event’s liquidation revenue. Engineering costs (\$500K–\$1M+ per solver) are additional.

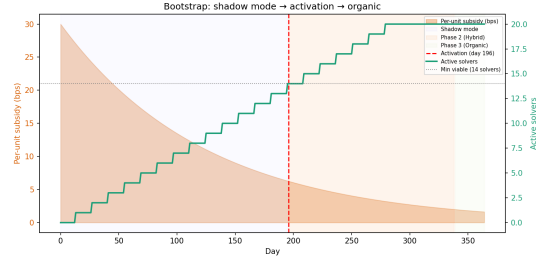


Figure 8: Shadow mode (days 0–196) → activation → self-sustaining (day 339).

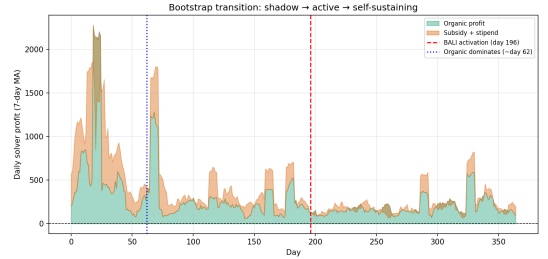


Figure 9: Organic vs subsidized profit. Standby stipend bridges the gap during shadow mode.

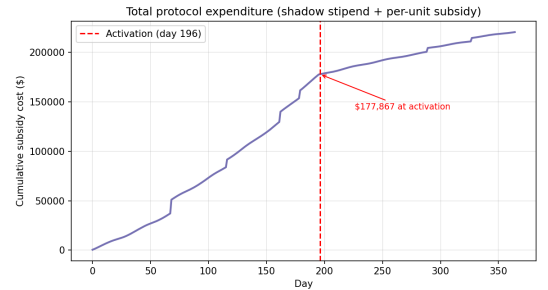


Figure 10: Cumulative expenditure with activation point annotated.

Table 5: Stress events (BTC/USD, Feb–Apr 2026).

Exchange	Date	DD	Peak	Trough
Hyperliquid	03-03	9.7%	\$73,651	\$66,541
dYdX	03-03	9.7%	\$73,692	\$66,564
Lighter	03-03	9.7%	\$73,644	\$66,523
dYdX	03-26	7.5%	\$70,887	\$65,603
Hyper.	03-26	7.4%	\$70,871	\$65,612
Lighter	03-05	7.4%	\$73,351	\$67,916

6 MULTI-EXCHANGE STRESS SIMULATION

6.1 Methodology

We test the architecture (Section 4) under stress using real price paths from three exchanges—Hyperliquid, dYdX, Lighter—over 60 days of hourly BTC/USD data (Feb–Apr 2026). Each exchange uses its calibrated λ from (Shah, 2026): Hyperliquid (3.29×10^{-5}), dYdX (3.74×10^{-5}), Lighter (3.19×10^{-5}).

Batch interval: 4 hours. Conservative; production would batch more frequently. Cascade depth: integer-valued (counts liquidation rounds). The “phase transition at 70%” is a step from $D=2$ to $D=1$: BALI absorbs enough that the residual never triggers a second round. A continuous cascade model might yield a non-sharp threshold. 80 paired runs at 6 coverage levels; 250 synthetic accounts, $3\text{--}25\times$ leverage, 70% correlation.

6.2 Stress Events

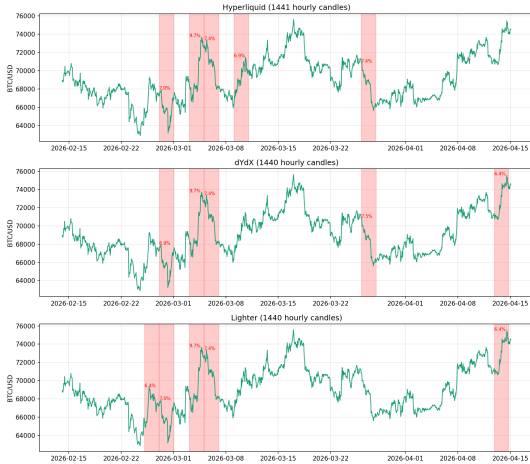
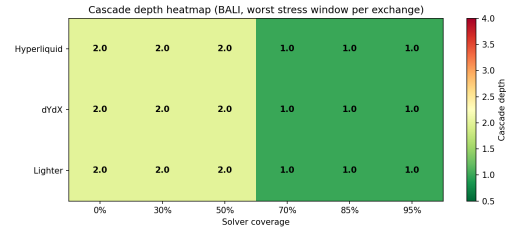
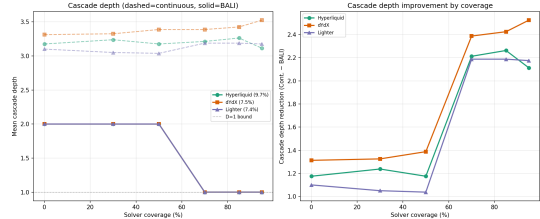


Figure 11: BTC/USD across three exchanges (60 days). Red = stress windows.

Table 6: At 70% coverage (80 paired runs). t : paired t -stat.

Exch.	Date	DD Cont.	D	BALI	BD	t
dYdX	03-03	9.7%	2.95	1.00	0	−20.4
Lighter	03-03	9.7%	2.92	1.00	0	−20.8
Hyper.	03-03	9.7%	3.17	1.00	0	−21.1
dYdX	03-26	7.5%	3.31	1.00	0	−26.6
Hyper.	03-26	7.4%	3.05	1.00	0	−24.3
Lighter	03-05	7.4%	3.10	1.00	0	−25.1

6.3 Results at 70% Coverage

Figure 12: Cascade depth heatmap: exchanges $\times$ coverage. $D=1$ at $\geq 70\%$.Figure 13: Cascade depth vs coverage. Step from $D=2$ to $D=1$ between 50% and 70%.

6.4 BALI Increases Bad Debt at Low Coverage

Below 50% coverage, BALI produces $20\text{--}100\times$ more bad debt than continuous liquidation (Table 7). The architecture’s coverage gate (Section 4) prevents this by design: batches form only when $c_{\text{eff}} \geq 0.70$.

The 70% sufficiency has a mechanical explanation: because impact is superlinear ($\alpha=1.5$), the 30% residual generates proportionally much less impact than its volume share. At 50%, the residual triggers a second cascade round—worst of both worlds.

Table 7: Bad debt at low coverage (Hyper., Mar 3, 80 paired runs).

Coverage	Cont. BD	BALI BD	Ratio	t
0%	\$0.24	\$4.85	20×	5.3
30%	\$0.09	\$9.23	103×	6.4
50%	\$0.15	\$12.63	84×	6.2
70%	\$0.00	\$0.00	—	−1.0

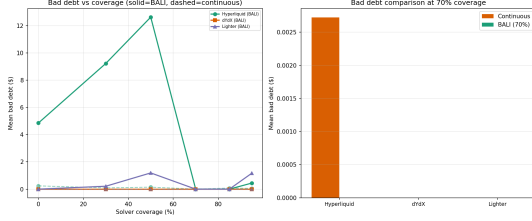


Figure 14: Bad debt vs coverage. BALI exceeds continuous at 0–50%, zero at 70%+.

7 DISCUSSION AND CONCLUSIONS

7.1 Summary

1. Participation conditions are computable under dual-channel adverse selection ($f_{\text{tox}}^* = 0.24$ at $\$50\text{K}/\sigma = 0.03$, robust to $\beta \in [2, 8]$), stress slippage ($\phi = 2.5$), and 2-hour inventory risk. Solver economics are strong during stress ($\geq \$195/\text{solver}$) but marginal between events.

2. The architecture resolves structural problems. The coverage gate prevents premature activation. Sealed-bid clearing with concentration caps ensures solver diversity. The solver registry enforces diversification ($p_c \leq 0.15$). Undersubscription cancellation prevents the low-coverage harm.

3. Bootstrapping costs $\sim \$220\text{K}$ in a two-phase model: 196 days shadow mode ($\$178\text{K}$), then activation. The coverage gate makes activation automatic, not discretionary.

4. Coverage is binary. 70%+ eliminates cascades ($t = -21.1$); below 50% increases bad debt 20–100×. Correlated default requires 88% nominal (at $p_c = 0.15$) for 70% effective.

7.2 Attack Surfaces

The architecture introduces specific attack surfaces for security analysis:

- Oracle manipulation: An attacker who can suppress the drawdown signal prevents batch formation during a real crash. Mitigation: multiple redundant price sources.

- Solver collusion: Solvers could coordinate to bid only at high discounts, extracting value from the exchange. Mitigation: sealed bids + the fallback to continuous liquidation caps the extractable discount.
- Heartbeat spoofing: A solver could maintain heartbeat while being operationally offline, inflating c_{eff} . Mitigation: bond slashing on non-delivery.
- Coverage manipulation: An attacker controlling multiple solver slots could withdraw simultaneously to force fallback. Mitigation: diversification requirements + staggered unbonding period.

7.3 Limitations

1. BTC/USD only. The architecture assumes solvers can hedge. For illiquid altcoins (e.g., JELLY), the hedging assumption breaks entirely; participation threshold may be infinite.
2. Synthetic positions. The “70%” threshold is approximate; real distributions may shift it.
3. Integer cascade depth. Cascade rounds are discrete; a continuous model might yield a non-sharp threshold.
4. No formal security proof. The architecture is specified but not formally verified. Incentive compatibility of the sealed-bid mechanism is assumed, not proved.
5. Development costs. $\$220\text{K}$ covers protocol subsidies, not $\$500\text{K}$ – $\$1\text{M}+$ per-solver engineering.

7.4 Open Problems

- Incentive-compatible auction design: proving truthful bidding is dominant under the proposed clearing rule.
- Optimal batch interval: balancing cascade prevention against batching delay.
- Multi-asset extension: adapting the architecture for assets where hedging venues are thin or nonexistent.
- Formal verification: proving safety properties (“coverage gate prevents harm”) under Byzantine solver behavior.

7.5 Concluding Remarks

This paper contributes a concrete architecture and economic analysis for solver-backed batch

liquidation. The mechanism’s binary nature—helpful above 70%, harmful below 50%—makes deployment a threshold problem, not a gradual optimization. The architecture’s coverage gate resolves this structurally: BALI cannot activate until solvers are ready, and reverts automatically if coverage drops mid-crisis. The two-phase bootstrapping model funds solver readiness at \$220K over 12 months. Whether this investment is justified depends on the frequency of future cascade events—but the infrastructure gap that BALI addresses is real, as the October 2025 crash demonstrated at \$10B+ scale.

ACKNOWLEDGEMENTS

AI tools were used to assist with code development and editorial refinement. All research design, analysis, and technical contributions are the author’s own.

REFERENCES

- CoinDesk (2025). HyperLiquid delists JELLY after vault squeezed in \$13m tussle. <https://coindesk.com/markets/2025/03/26/hyperliquid-delists-jellyjelly>.
- CoinDesk Research (2025). Crypto’s black friday: \$19 billion liquidation cascade. <https://coindesk.com/coindesk-indices/2025/10/15/crypto-s-black-friday>.
- CoW Protocol (2024). Fair combinatorial batch auction. <https://docs.cow.fi/overview/batch-auctions>.
- dYdX Foundation (2024). dYdX v4: Decentralized perpetual exchange on cosmos. <https://dydx.exchange>.
- dYdX Foundation (2025). October 2025: dYdX chain incident review. <https://dydx.xyz/blog/october-2025-dydx-chain-incident-review>.
- Enrico, E. (2025). MEV liquidation: Phase a analysis. GitHub repository.
- Hyperliquid Labs (2024). Hyperliquid: Onchain perpetual futures exchange. <https://hyperliquid.xyz>.
- Kulkarni, A. et al. (2025). Autodeleveraging: Impossibilities and optimization. arXiv preprint arXiv:2512.01112.
- Kulkarni, A. et al. (2026). Optimal online autodeleveraging. arXiv preprint arXiv:2602.15182.
- MakerDAO (2020). Black thursday post-mortem. <https://blog.makerdao.com>. March 2020.
- Perez, D., Werner, S., Xu, J., and Livshits, B. (2021). Liquidations: DeFi on a knife-edge. Financial Cryptography and Data Security (FC 2021), pages 457–476.
- Qin, K., Zhou, L., Livshits, B., and Gervais, A. (2021). Attacking the DeFi ecosystem with flash loans for fun and profit. In Financial Cryptography and Data Security (FC 2021), pages 3–32.
- Shah, N. (2026). BALI: Solvency-constrained batch liquidation as a cryptoeconomic control mechanism. In Proc. SECRIPT 2026 (submitted).